Optimal Reconfiguration of Systems in Aeronautics

Lecture 09: “Linear Optimal Control and Progressive Fault Accommodation”

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Outline

1. Classical vs. Progressive Fault Accommodation
2. B747 Application Example

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1. Classical vs. Progressive Fault Accommodation
   - Classical Accommodation Scheme
   - Drawbacks of the Classical Accommodation
   - Progressive Accommodation Scheme
   - Critique on Progressive Accommodation
2. B747 Application Example

Classical accommodation scheme I

Assume that N is the number of steps needed to obtain a satisfactory approximation for the solution to the post-fault CARE using the Newton-Kleinman or Matrix-Sign-Function approaches; denote by $\bar{\Delta}_k$ the duration of iteration k.

A classical accommodation procedure would naturally apply the nominal control law (1), respectively

$$u_n(t) = -K_n x(t) \tag{1}$$

with

$$K_n = R_n^{-1} B_n^\top X_n \,, \tag{2}$$

during the period $t_{FTC} - t_{FDIE} = \bar{\Delta}$ $\left(\bar{\Delta} = \sum_{k=1}^{N} \bar{\Delta}_k \text{ and } \bar{\Delta}_k = \sum_{l=1}^{k} \Delta_l\right)$ until the accommodated feedback is available, where t_{FDIE} is the time at which the new post-fault pair (A_f, B_f) is given by the FDIE module, while t_{FTC} is the instant at which the new post-fault control matrix K_f computed by the FAR module becomes available (assimilate $\bar{\Delta}$ with the duration to obtain the solution to CARE$_f$ using the Schur Vector Method, as well).

Classical accommodation scheme II

Hence, using the Classical Accommodation Scheme results in a two-stage post-fault behavior, as sketched in the following algorithm.

Classical Accommodation Scheme (CAS)

(1-1) While computing the accommodated control law,

$$\dot{x}(t) = (A_f - B_f K_n)x(t)\,, \text{ for } t \in [t_f, t_{FDIE} + \bar{\Delta})\,.$$

(1-2) When iterations reached convergence,

$$\dot{x}(t) = (A_f - B_f K_{f_N})x(t)\,, \text{ for } t \in [t_{FDIE} + \bar{\Delta}, +\infty)\,,$$

with $N \in \{N_{NK}, N_{MSF}\}$. $\square$

Drawbacks of the classical accommodation I

For the situation described in the classical accommodation scheme, the computation delay is $t_{FTC} - t_{FDIE} = \bar{\Delta}$, but note also that the occurrence of a fault differentiates four periods in the system operation, namely:

1. fault-free - $t \in [0, t_f)$: the system is in normal operation, model M_n - the applied control is the nominal one, u_n.
2. fault detection - $t \in [t_f, t_{FDIE})$: the system is faulty, model M_f, but the FDIE algorithm has not detected, isolated, and identified / estimated the fault yet and the control has not been accommodated - the nominal control u_n is still applied.
3. recovery control computation - $t \in [t_{FDIE}, t_{FTC})$: the system is faulty, model M_f, the FDIE algorithm has detected, isolated, and identified / estimated the fault, but the control has not been accommodated / reconfigured yet - the nominal control u_n is still applied.
4. fault-recovered - $t \in [t_{FTC}, +\infty)$: the system is faulty, model M_f - the accommodated / reconfigured control u_f has been computed and applied.

Drawbacks of the classical accommodation II

Hence, the total delay, $t_{FTC} - t_f$, is the sum of the fault detection, isolation, and identification / estimation delay, $t_{FDIE} - t_f$, and of the post-fault control accommodation delay, $t_{FTC} - t_{FDIE}$. Obviously, the feedback controlling the post-fault system is inappropriate on the time interval $[t_f, t_{FTC})$. Thus, the major drawback of CAS is that, during the computation time of the solution to CARE$_f$, the closed-loop stability and performance are uncertain since the post-fault system is under the nominal control.

Forth, the deviation during the transient trajectory might be such that, although stabilizing the system after t_{FTC}, the accommodated control might lead to a non-admissible value of the cost $J_{M_f}(x_f, u_f)$, i.e., condition (3), respectively

$$J_f \leq \rho J^* \,, \tag{3}$$

with

$$\rho = \frac{J_{\max}}{J^*} \geq 1 \,,$$

might not hold.

Therefore, the FTC delay, $t_{FTC} - t_{FDIE}$, which adds to the FDIE delay, $t_{FDIE} - t_f$, might imply the non-admissibility of the accommodated control and/or the system loss; as minimizing the quantity $t_{FDIE} - t_f$ is the objective of FDIE research, shortening the period of the post-fault control accommodation is desired in the FTC setting.

Progressive accommodation scheme I

On the contrary to the classical scheme, the progressive accommodation approach would apply the control laws

$$u_{f_k}(t) = -K_{f_k} x(t)\,, \text{ for } k = 1, 2, \ldots, N\,, \text{ with } N \in \{N_{NK}, N_{MSF}\}\,,$$

with K_{f_k}'s computed upon the solutions X_{f_k} provided either by (4), respectively

$$\begin{aligned} K_{f_k} &= R_n^{-1} B_f^\top X_{f_{k-1}}\,, \\ \hat{A}_{f_k} &= A_f - B_f K_{f_k}\,, \end{aligned} \tag{4}$$

for $k =_{k\to+\infty} N_{NK}$ and K_{f_0} chosen such that the initial closed-loop matrix is stable, that is $\hat{A}_{f_0} = A_f - B_f K_{f_0}$ has eigenvalues with negative real-parts, or (5), respectively

$$\begin{bmatrix} & Z_{f_{k_{12}}} \\ Z_{f_{k_{22}}} & + I_n \end{bmatrix} X_{f_k} = -\begin{bmatrix} Z_{f_{k_{11}}} & + I_n \\ Z_{f_{k_{21}}} & \end{bmatrix}\,, \tag{5}$$

for $k =_{k\to+\infty} N_{MSF}$, after partitioning $Z_{f_{N_{MSF}}}$ accordingly, rather than waiting for the complete convergence of the iterations.

Progressive accommodation scheme II

Hence, using the Progressive Accommodation Scheme results in a three-stage post-fault behavior, as sketched in the following algorithm.

Progressive Accommodation Scheme (PAS)

(5-1) During the first iteration,

$$\dot{x}(t) = (A_f - B_f K_n)x(t)\,, \text{ for } t \in [t_f, t_{FDIE} + \Delta_1)\,.$$

(5-2) While progressive accommodation proceeds,

$$\dot{x}(t) = (A_f - B_f K_{f_k})x(t)\,, \text{ for } t \in [t_{FDIE} + \bar{\Delta}_k, t_{FDIE} + \bar{\Delta}_{k+1})\,.$$

(5-3) When iterations reached convergence,

$$\dot{x}(t) = (A_f - B_f K_{f_N})x(t)\,, \text{ for } t \in [t_{FDIE} + \bar{\Delta}, +\infty)\,,$$

with $N \in \{N_{NK}, N_{MSF}\}$. $\square$

Monotonic convergence of the Newton-Kleinman approach I

Remember that $X_{f_1} \leq X_{f_0}$ is not always true in the Newton-Kleinman method.

With respect to the ordering (6), respectively

$$0 \leq X_{f_{N_{NK}}} \leq \ldots \leq X_{f_{k+1}} \leq X_{f_k} \leq \ldots \leq X_{f_1} \,, \tag{6}$$

the monotonic convergence from the beginning is conditioned by the initial step.

Introduce $\left\| \hat{\Omega}_f^{-1} \right\|_2$ the operator-norm of the inverse closed-loop Lyapunov operator, i.e.,

$$\hat{\Omega}_f(Y_f) \equiv (A_f - B_f B_f^\top X_{f_{N_{NK}}})^\top Y_f + Y_f (A_f - B_f B_f^\top X_{f_{N_{NK}}}) \,,$$

computed upon the solution to

$$(A_f - B_f B_f^\top X_{f_{N_{NK}}})^\top Y_f + Y_f (A_f - B_f B_f^\top X_{f_{N_{NK}}}) + Q_n = 0 \,,$$

where the operator-norm is defined as

$$\left\| \hat{\Omega}_f \right\|_2 = \max_{Y_f \in \mathbb{R}^{n \times n}_{Y_f \neq 0}} \frac{\left\| \hat{\Omega}_f(Y_f) \right\|_2}{\| Y_f \|_2} \,;$$

Monotonic convergence of the Newton-Kleinman approach II

for simplicity, the control weighting matrix R_n has been taken equal to unity.

Forth, consider $r > 0$ the supremum over all radii of the balls $\mathscr{B}_r(X_{f_{N_{NK}}})$ centered at $X_{f_{N_{NK}}}$ such that, if $X_{f_0} \in \mathscr{B}_r(X_{f_N})$, where

$$r \equiv \frac{1}{3\|B_f\|_2^2 \left\|\hat{\Omega}_f^{-1}\right\|_2} > 0\,,$$

then X_{f_1} satisfies (7), respectively

$$\hat{A}_{f_k}^\top X_{f_k} + X_{f_k}\hat{A}_{f_k} + K_{f_k}^\top R_n K_{f_k} + Q_n = 0\,, \tag{7}$$

for $k = 1$, i.e.,

$$\hat{A}_{f_1}^\top X_{f_1} + X_{f_1}\hat{A}_{f_1} + K_{f_1}^\top K_{f_1} + Q_n = 0\,,$$

and $\left\|X_{f_{N_{NK}}} - X_{f_1}\right\|_2 \leq \left\|X_{f_{N_{NK}}} - X_{f_0}\right\|_2$, with equality only when $X_{f_0} = X_{f_{N_{NK}}}$, that is whether the initial guess is identical with the solution.

Monotonic convergence of the Newton-Kleinman approach III

Hence, after the fault changes the pair (A_n, B_n) to (A_f, B_f) and the initial step for the Newton-Kleinman iteration is chosen so that $\left\| X_{f_{N_{NK}}} - X_{f_0} \right\|_2 < r$ does not hold, there is a non-increasing monotonic series of solutions to the post-fault CARE after the first step only.

However, after the first iteration, this method guarantees improving the control solution in terms of the value of the quadratic cost, that is it can be called an "anytime" algorithm.

Anytime property of the Newton-Kleinman method I

Obviously, in real-time operation, there are cases when the computation time needed to obtain precise or optimal solutions usually reduce the overall utility of the control system, which might become critical in certain failure conditions. That conducts to the concept of anytime algorithms, which are algorithms whose quality of results improves gradually as computation time increases.

The special feature of anytime algorithms is the use of well-defined quality measures to monitor the progress in problem solving and allocate computational resources effectively. With respect to the anytime algorithms construction, three metrics have proved to be useful, namely certainty (measures the degree of certainty that the result is correct), accuracy (measures the degree of accuracy or how close the approximate result is to the exact answer), and specificity (measures the level of detail of the result).

Anytime property of the Newton-Kleinman method II

As not every algorithm that produces a series of approximate answers is a well-behaved anytime algorithm, a set of desired properties include the following:

- measurable quality: the quality of an approximate result can be determined precisely;
- recognizable quality: the quality of an approximate result can easily be determined at run time within a constant time;
- monotonicity: the quality of the result is a nondecreasing function of time and input quality;
- consistency: the quality of the results is correlated with the computation time and input quality;
- diminishing results: the improvement in solution quality is bigger at the early stages of the computation and diminishes over time;
- interruptibility: the algorithm can be stopped at any time and provide some answer;
- preemptability: the algorithm can be suspended and resumed with minimal overhead.

Anytime property of the Newton-Kleinman method III

In this respect, for the Newton-Kleinman approach used in progressive fault accommodation, the metric that would be naturally imposed is the accuracy, measuring the distance between the solution to the intermediate continuous algebraic Lyapunov equations and the solution to the classical continuous algebraic Riccati equation.

Specifically, the solutions obtained with this technique are inputs to the result that interests the most, namely the cost of the post-fault accommodation, which is a measurable quality and also recognizable at the end of every sample $\bar{\Delta}_k$, provided the evaluation of the quadratic integral criterion.

Also, the cost decreases over time and thus the quality of the control solution is nondecreasing as it is cheaper or at least equal to the previous one at every step, that being guaranteed by the monotonicity of the solutions of the series of continuous algebraic Lyapunov equations. Moreover, the quadratic convergence rate provided by the Newton method is in accordance with the diminishing results property.

Finally, the approach is preemptive as long as it achieves stability after the first step provided the initial stabilization procedure and resuming it can be done using the control gain matrix obtained during the last iteration.

Newton-Kleinman progressive accommodation and step-size-control I

Unfortunately, given any arbitrary starting point, the Newton-Kleinman approach method may take several steps before entering the region of quadratic convergence. Nevertheless, the initial slow convergence may be accelerated through step-size-control by using an exact-line-search technique.

Usually, after the initial starting "guess" is available, the accommodation of the solution to CARE$_f$ is done as follows

$$X_{f_{k+1}} = X_{f_k} + W_{f_k} \,,$$

where W_{f_k} represents the solution to the continuous algebraic Lyapunov equation

$$(A_f - G_f X_{f_k})^\top W_{f_k} + W_{f_k}(A_f - G_f X_{f_k}) = R_{f_k} \,,$$

for the residual

$$R_{f_k} = -R(X_{f_k}) = -(A_f^\top X_{f_k} + X_{f_k} A_f - X_{f_k} G_f X_{f_k} + Q_n) \,. \tag{8}$$

Newton-Kleinman progressive accommodation and step-size-control II

Improving the accommodation procedure with an exact-line-search may therefore be expressed as

$$X_{f_{k+1}} = X_{f_k} + t_{f_k} W_{f_k} \,,$$

where t_{f_k} is an "exact" minimizer of the Frobenius-norm of the next residual $R_{f_{k+1}}$ computed as in (8).

Concerning this aspect, the suggested appropriate interval for the step-size is $t_{f_k} \in (0, 2]$ and recommended values are from its first-half for the first steps and from the second-half for the rest of the process.

However, endowing the progressive accommodation with step-size-control does not ensure the monotonic convergence of the Newton-Kleinman approach, that is the positive-definiteness of iterates X_{f_k} is not guaranteed anymore.

Hammarling initial step for the Newton-Kleinman approach I

Return to finding the initial "guess" in Bass-Armstrong approach, that is solving (9), respectively

$$\bar{A}_f W_f + W_f \bar{A}_f^\top + 2 B_f B_f^\top = 0\,, \tag{9}$$

with the Bartels-Stewart algorithm, i.e., finding the solution to the Lyapunov equation

$$\bar{A}_f W_f + W_f \bar{A}_f^\top + 2 B_f B_f^\top = 0\,, \tag{10}$$

and note that, if $\bar{A}_f$ is stable and the quantity given by $B_f B_f^\top$ is positive-definite, then nonnegative-definite is also the matrix W_f; consider then its Cholesky factorization, $W_f = \tilde{W}_f \tilde{W}_f^\top$, and remember that the numerical conditioning of W_f w.r.t. inversion is given by

$$\kappa_f(W_f) = \|W_f\|_S \left\| W_f^{-1} \right\|_S ,$$

for which it also holds that

$$\kappa_f(W_f) = \kappa_f^2(\tilde{W}_f)\,,$$

where $\|.\|_S$ denotes the spectral-norm, i.e., the maximum singular value of W_f, and therefore W_f may be more ill-conditioned w.r.t. inversion than its factor $\tilde{W}_f$; hence, it gets desirable to obtain a more accurate post-fault control matrix $K_f = B_f^\top W_f^{-1}$ with the help of the Cholesky factors $\tilde{W}_f$ instead of using the modified Bartels-Stewart algorithm, that is solving (10) directly for the Cholesky factors using the Hammarling approach.

Hammarling initial step for the Newton-Kleinman approach II

Unfortunately, it is assumed that the quantity $B_f B_f^\top$ is positive-definite, but note that, if B_f models complete actuators faults, then their corresponding columns will become zero, that is B_f gets rank-deficient and invalidates the algorithm.

At any rate, numerical improvements of the entire stabilization process are obtained when approaching the initial step problem with the method of Bass-Armstrong with the RSF of A_f, that is

$$\left[-(\check{A}_f + \beta_f I_n)\right] \check{W}_f + \check{W}_f \left[-(\check{A}_f + \beta_f I_n)\right]^\top + 2\check{C}_f = 0\,,$$

for $\check{A}_f = U_f^\top A_f U_f$, $\check{W}_f = U_f^\top W_f U_f$, $\check{B}_f = U_f^\top B_f$ and $\check{C}_f = \check{B}_f \check{B}_f^\top$, the reduction with $3n^3 - 2n^2 m$ flops in the computational effort of Bartels-Stewart algorithm being eventually reported in this case.

Increasing the solution accuracy with defect correction I

Due to finite arithmetic computations, the solution to CARE$_f$ is clearly an approximate one, denote it by $\hat{X}_f$.

However, the deviation from the "perfect" one, $\tilde{X}_f = X_f - \hat{X}_f$, solves again a Riccati-type equation

$$\hat{A}_f^\top \tilde{X}_f + \tilde{X}_f \hat{A}_f - \tilde{X}_f B_f R_n^{-1} B_f^\top \tilde{X}_f + R_f(\hat{X}_f) = 0\,,$$

where $\hat{A}_f = A_f - B_f R_n^{-1} B_f^\top \hat{X}_f$.

Furthermore, the approximate solution $\hat{X}_f$ is the exact solution of

$$A_f^\top \hat{X}_f + \hat{X}_f A_f - \hat{X}_f B_f R_n^{-1} B_f^\top \hat{X}_f + \left[Q_n - R_f(\hat{X}_f)\right] = 0\,.$$

Since both procedures implementing the progressive accommodation need applying 1-2 steps of the defect-correction algorithm to improve the accuracy of the computed solution to the post-fault CARE, it does not make any difference between the Newton-Kleinman and the Matrix-Sign-Function approaches in terms of number of steps. However, Matrix-Sign-Function is used sometimes as a refinement method for the solution obtained with Newton-Kleinman.

Numerical conditioning of the accommodated solution I

Whether actuator faults of small magnitude only are provided to happen, i.e., there exists a small ϵ such that $\|B_n - B_f\|_2 < \epsilon$, then CARE$_n$ and CARE$_f$ are said to be similar.

In this perspective, the numerical conditioning of the post-fault CARE could be investigated w.r.t. the nominal one

$$\kappa_f(X_f) = \lim_{\epsilon \to 0} \frac{\|X_n - X_f\|_2}{\|X_f\|_2}\,, \text{ for } \|B_n - B_f\|_2 < \epsilon\,.$$

Note however that, if actuator and system faults are both considered, then the difference between the norms of the nominal and post-fault system respectively control matrices can be at least one order of magnitude far one from each other such that there is no single small ϵ to prove the similarity of the nominal and post-fault CARE's and therefore two norm limits ϵ_A and ϵ_B must be taken for the two different types of faults.

Numerical conditioning of the accommodated solution II

Then, analyzing the numerical conditioning based on

$$\kappa_f(X_f) = \lim_{\epsilon_{A,B}\to 0} \frac{\|X_n - X_f\|_2}{\|X_f\|_2}\,, \text{ for } \|A_n - A_f\|_2 < \epsilon_A \text{ and } \|B_n - B_f\|_2 < \epsilon_B\,,$$

becomes much more complicated if $\epsilon_A \nsim \epsilon_B$; also, the well-posedness of CARE$_f$ is questionable as it is basically dependent on the correctness with which the pair (A_f, B_f) has been re-estimated by an FDIE algorithm (care must be exercised to avoid any confusion in the perception of rounding errors and influence of faults).

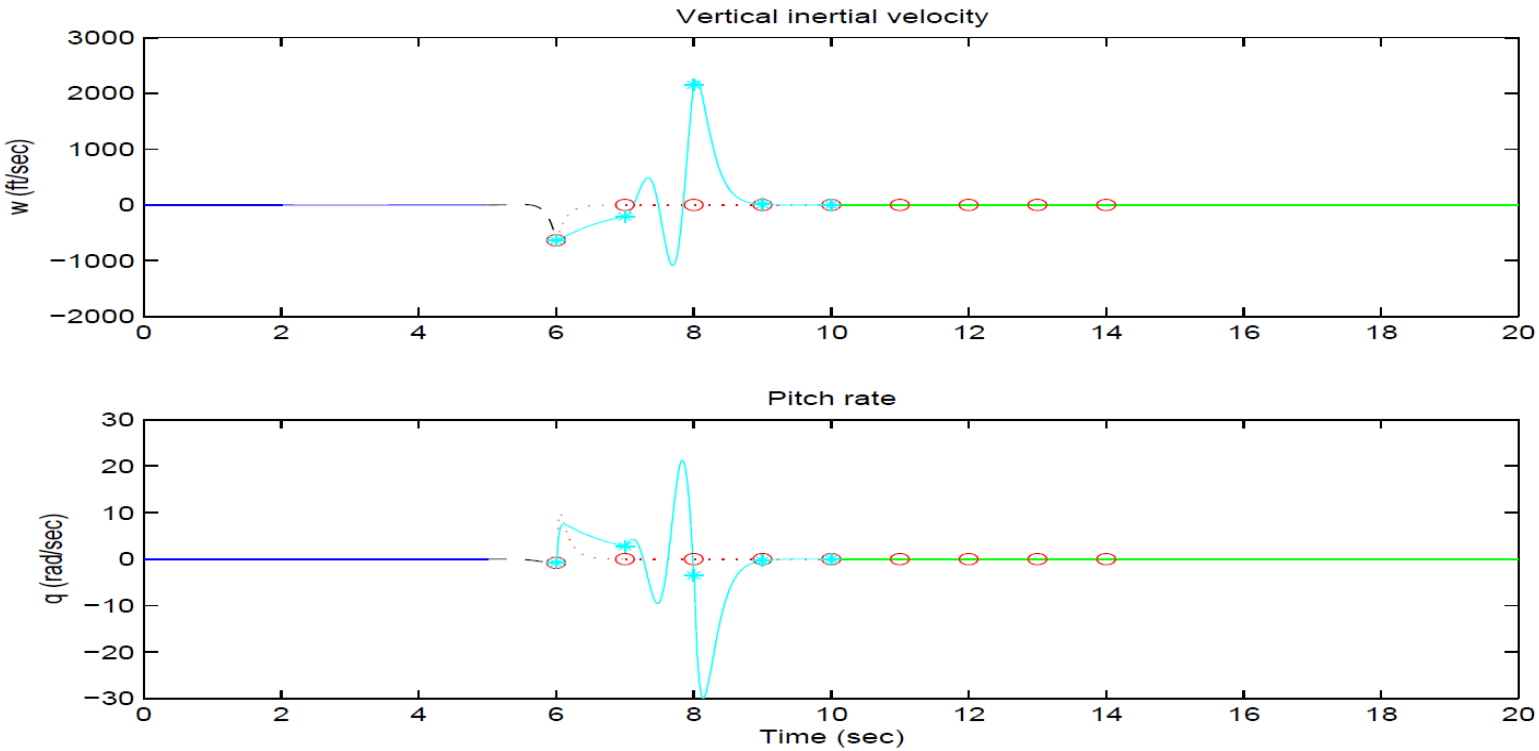
In the sequel, an application example is developed to prove the features of the classical and progressive accommodation schemes when both are used to design a fault tolerant control sequence that addresses the structural fault in the model of the short-period mode of a B747 aircraft respectively the loss-of-efficiency in the elevator actuator.

Outline

1. Classical vs. Progressive Fault Accommodation
2. B747 Application Example
   - Closed-Loop Simulation
   - Evaluate the Cost Functional

Closed-loop simulation I

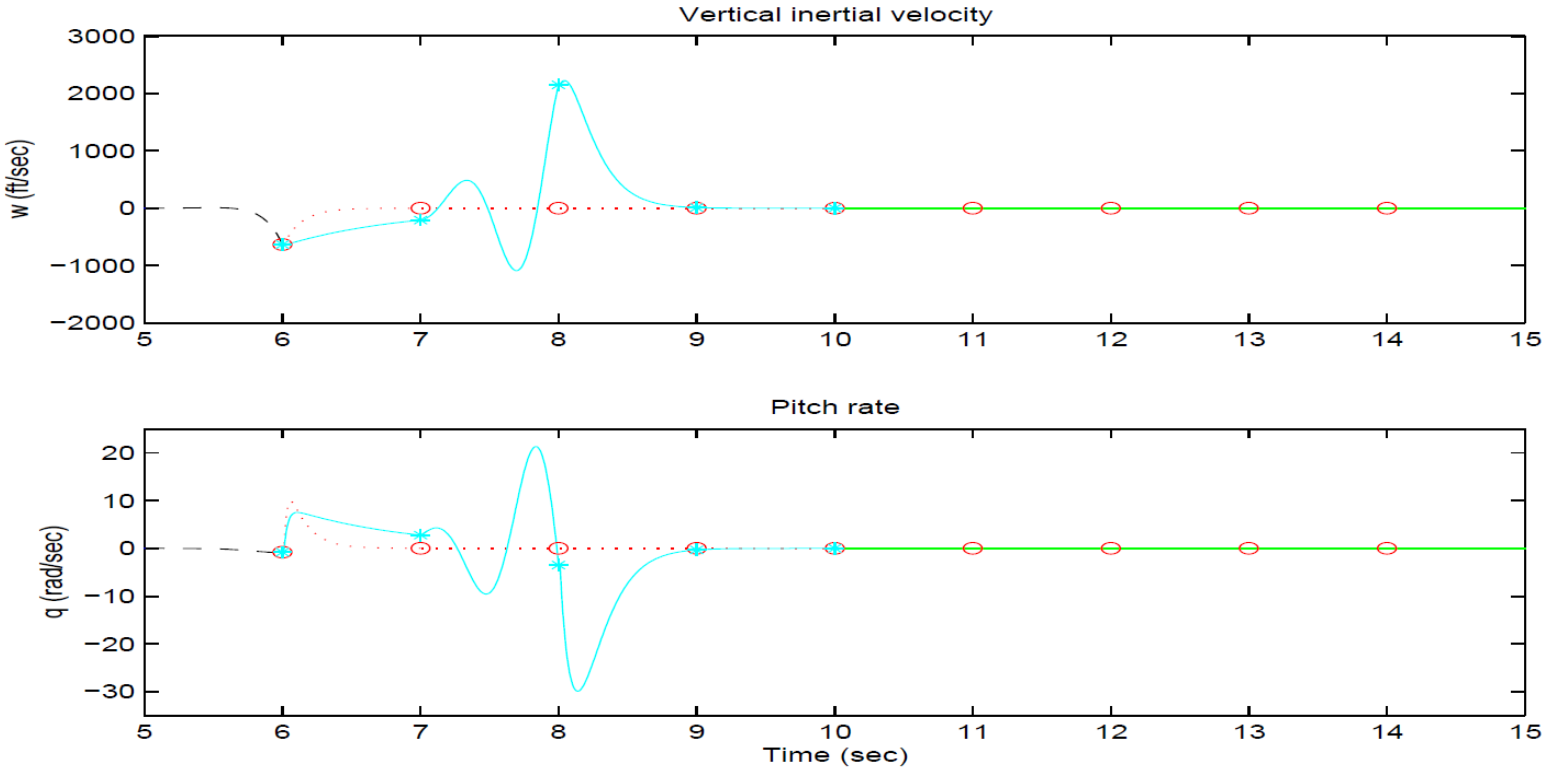
The following figure displays the simulations of the closed-loop behavior during the post-fault period when applying the progressive and the classical control schemes. The fault occurred at $t_f = 4$ *sec*, the estimation of the faulty model was ready at $t_{FDIE} = 5$ *sec*, therefore there is one sample of uncertain evolution, until $t_{FDIE+\Delta} = 6$ *sec*, when the solution to the initial step problem will be readily available, this one being applied until $t_{FDIE+2\Delta} = 7$ *sec*.



NK- / MSF- PAS vs. CAS for B747-SP - Unstable Case (Continuous)

Closed-loop simulation II

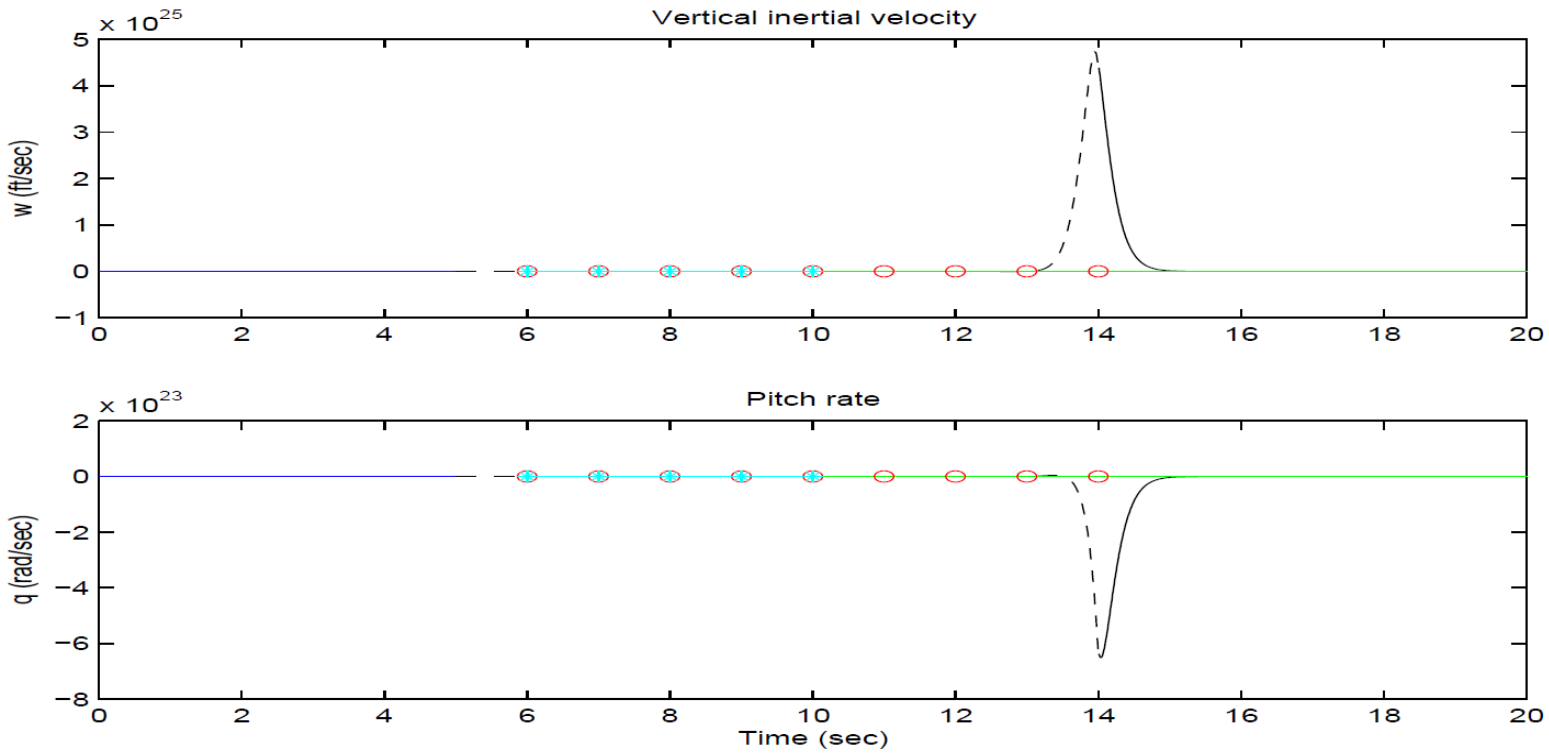
The following figure highlights the central graphs only.



NK- / MSF- PAS vs. CAS for B747-SP - Unstable Case (Transient)

Closed-loop simulation III

Also, see in the following figure the huge overshoot that is encountered if classical accommodation procedure is to be used afterwards convergence only.



CAS for B747-SP - Unstable Case (Continuous)

The closed-loop instability boils down during the convergence process such that the recovery graphs make the ones showing the progressive accommodation appear insignificant. (Compare the w- and q- scales in the first and the third of the previous figures, respectively.)

Evaluate the cost functional I

At $t_f = 6\ sec$, the post-fault "initial" condition estimates as

$$x_f = \begin{bmatrix} -632.6166 & -0.8096 \end{bmatrix}^\top ;$$

therefore, using the solution to the accommodated CARE$_f$, that is

$$X_f^{acc} = \begin{bmatrix} 0.0004 & 0.0149 \\ 0.0149 & 1.6434 \end{bmatrix} \times 10^3 , \tag{11}$$

it yields a value of

$$J_{M_f}(x_f, u_f) = 1.7813 \times 10^5 ,$$

for the quadratic cost (12), respectively

$$J_{M_f}(x_f, u_f) \triangleq J_f = x_0^\top X_n x_0 + x_f^\top (X_f - X_n) x_f . \tag{12}$$

Evaluate the cost functional II

Hence, evaluating the condition in (3) provides the magnitude

$$\rho = 1.4425 \times 10^8 \,,$$

that is any upper cost limit fixed by

$$J_{\max} \leq 1.7813 \times 10^5$$

would invalidate the accommodated control law.

Conclusion I

In these series of lectures, the nominal and fault tolerant linear quadratic control problems are described.

Upon direct and iterative methods to solving the post-fault Continuous Algebraic Riccati Equation, two fault accommodation strategies are proposed in order to obtain the optimal post-fault feedback, namely the classical and progressive accommodation schemes. An application to the accommodation for the structural fault in the short-period mode model of an aircraft and the loss-of-effectiveness of the actuator controlling the elevator deflection was also developed. Simulations reveal the fact that the progressive accommodation strategies enhance the post-fault closed-loop behavior.

Moreover, among the two methods, the Newton-Kleinman approach is the one recommended, as it improves the optimal cost at every iteration albeit requiring the initial step being stabilizable, while the Matrix-Sign-Function does not possess such an "anytime" property.

Questions? vs. Answers!

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